Queensland University of Technology
Brisbane Australia

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A Decentralised Bilateral Energy Trading System for Peer-to-Peer Electricity Markets

Mohsen Khorasany, *Student Member, IEEE*, Yateendra Mishra, *Member, IEEE*, and Gerard Ledwich, *Senior Member, IEEE*

***Abstract*— Increase in the deployment of Distributed Energy Resources (DERs) has triggered a new trend to redesign electricity markets as consumer-centric markets relying on Peer-to-Peer (P2P) approaches. In the P2P markets, players can directly negotiate under bilateral energy trading to match demand and supply. The trading scheme should be designed adequately to incentivise players to participate in the trading process actively. This paper proposes a decentralised P2P energy trading scheme for electricity markets with high penetration of DERs. A novel algorithm using primal-dual gradient method is described to clear the market in a fully decentralised manner without interaction of any central entity. Also, to incorporate technical constraints in the energy trading, line flow constraints are modelled in the bilateral energy trading to avoid overloaded or congested lines in the system. This market structure respects market players' preferences by allowing bilateral energy trading with product differentiation. The performance of the proposed method is evaluated using simulation studies, and it is found that market players can trade energy to maximise their welfare without violating line flow constraints. Also, compared with other similar methods for P2P trading, the proposed approach needs lower data exchange and has a faster convergence rate.**

***Index Terms*— Peer-to-Peer, Decartelised market, Bilateral trading, Product differentiation, Market framework design.**

I. Introduction

INCREASE in the deployment of Distributed Energy Resources (DERs) along with extending communication and control infrastructures to consumer level has motivated passive consumers to become active players by managing their generation and load [1]. The emergence of prosumers as new players has changed the traditional electricity markets to consumer-centric markets, where players can trade energy locally to manage their energy more efficiently. Appropriate design of consumer-centric markets can increase renewable energy integration in the distribution network via the adoption of a business model which provides direct interaction between prosumers. Peer-to-Peer (P2P) energy markets have been proposed as a new market design to incentivise local energy trading, where different forms of energy trading and market clearing mechanisms have been reported considering the characteristics of the competitive market [2].

The authors are with the School of Electrical Engineering and Computer Science, Queensland University of Technology, Brisbane, Queensland 4000, Australia (e-mail: m.khorasany@ieee.org; yateendra.mishra@qut.edu.au; g.ledwich@qut.edu.au).

The concept of P2P network has been extensively used for resource sharing in the field of computer science for distributed computation [3]. P2P transaction is an agreement between two peers to exchange a product or information. The primary objective of P2P trading is to encourage market players to trade with one another without any direct influence from the central controller. In the electricity market, a single P2P transaction between two players is no more than a single bilateral contract for exchanging an amount of energy for a specific price [4], where excess energy from various small-scale DERs is traded locally with players with energy deficits [5]. As players in the electricity market are part of the electricity network, they need to use this network for their trading, which imposes technical constraints on energy trading. In the decentralised P2P trading, incorporating network constraints in energy trading is of paramount importance, since neglecting technical constraints compromises operation of the network. Therefore, physical network constraints must be included in energy trading models.

Another important consideration is that the P2P market should respect market players' preferences by allowing bilateral energy trading with product differentiation. Product differentiation means that market players can express preferences on the type and quality of the energy they will exchange [6]. From a customer's perspective, such preference could be for power quality, level of reliability, time of use, the volume of usage, maximum demand, and level of environmental impact. Electricity is commonly treated as a homogeneous good, most often priced under uniform pricing in forward markets. However, due to recent transformative events in the electricity industry such as massive deployment of DERs, and technological advances metering, it has been found that expressing preferences should yield product differentiation and hence price differentiation [7, 8]. Applying product differentiation to electricity can significantly induce end-users to more effectively and efficiently satisfy their demands upon the system [9], however, the more product differentiation, the higher expected difficulty in finding a solution. Therefore, the appropriate P2P market design should allow product differentiation by making it possible to have different prices for different transactions, which implies that all market players may account for preferences, different valuations of electricity, and differentiated network charges through the market clearing process. Hence, the objective of this paper is to develop a decentralised bilateral trading method allowing product differentiation, while impact of each transaction on line flow constraints is considered.

For the market clearing in P2P trading, different optimisation methods can be used such as centralised optimisation, Alternating Direction Method of Multipliers (ADMM) [10], primal-dual gradient methods [11, 12], and consensus-based

methods [13, 14]. The computation overheads and privacy of centralised methods are of concern as they need to have information of all players. Hence, distributed methods have been proposed to address these concerns. ADMM method is a well-known method for distributed optimisation, where it generally needs a coordinator to generate a coordination signal. Authors in [15] use ADMM for market clearing, where community manager is the supervisor of convergence to optimality. ADMM is employed by authors in [10] for multi-class energy management, where a platform agent acts as a coordinator to set the price of each energy class. A platform for coordinating DER scheduling in microgrid using smart contracts in blockchain is presented in [16], where smart contracts serve as ADMM coordinator to update global variables. Using a supervisory node or coordinator means that the optimisation method is not fully decentralised.

This paper proposes a primal-dual gradient method which does not require a coordinator and can be implemented in a fully decentralised manner. Furthermore, the proposed algorithm is different from conventional primal-dual gradient algorithms as it uses the first-order method instead of interior-point method to solve sub-problems, which reduces computation per iteration time. Comparing with Consensus + Innovation method [13, 14], the proposed method needs lower information exchange for market clearing, which reduces convergence time and the required number of iterations. Authors in [14] use Relaxed Consensus+ Innovation method (RCI), where each player exchange both price and amount of energy in every iteration. However, the proposed algorithm is designed such that sellers only need to send their prices to the buyers, and buyers should respond to the sellers by updating their demands. The other advantage of the proposed method is that compared with Consensus + Innovation method, the convergence of the algorithm has lower sensitivity to the tuning parameters, and reaching convergence to the optimal solution can be guaranteed using a range of different tuning parameters. Also, different from [13] and [14], in this paper network constraints are considered in the energy trading.

Given this context, this paper proposes a fully decentralised bilateral trading scheme for P2P energy trading among market players. A decentralised market clearing algorithm is developed to clear the market for all peers, while line flow constraints are taken into account. This decentralised implementation offers benefits such as transparency, data privacy, and absence of a central supervisory node. The primal-dual gradient method is adopted for market clearing in multi-bilateral trading. In summary, the contributions of this paper are as follows:

- A novel bilateral trading scheme is proposed for direct interaction of players in the P2P market.
- A decentralised algorithm is developed for market clearing, which allows product differentiation and needs lower number of iteration and time for convergence compared with other market clearing methods.
- Line flow constraints are monitored during energy exchange, where Power Transfer Distribution Factor (PTDF) is employed to determine the contribution of each transaction in the line flow.
- The implementation of the proposed method is demonstrated to show the feasibility and effectiveness of this method for P2P energy trading.

The remainder of the paper is organised as follows. Section II presents related works. In Section III, the trading scheme is described, and optimisation problem for maximising the welfare of market players is formulated. Market clearing mechanism and executed algorithms by market players are discussed in Section IV. Numerical analysis, case studies, and discussion of the proposed method are presented in Section V, followed by conclusions in Section VI.

II. Related works

There is an increasing interest in the literature on the design of market scheme for P2P energy trading considering some of the aforementioned features. The related works are divided into two main threads. The first thread is concerned about market design for decentralised energy trading. A four-layered structure for P2P energy trading is proposed in [17], where energy trading is simulated using game theory. Authors in [18] propose a blockchain based P2P market design for local energy trading in microgrid. The decentralised market design proposed in [19] allows system operator to manage local demand by obtaining flexibility from prosumers and aggregators. Real-time P2P energy trading in a prosumer based community microgrid is modelled by authors in [20], where a game theoretic approach is employed to model interaction between sellers and buyers. A bilateral contract network for P2P energy trading between generators, suppliers, and prosumers is proposed in [21]. However, in these works constraints imposed by the network are neglected. Also, product differentiation in bilateral trading is not implemented. Authors in [10] propose a P2P market platform based on multi-class management, where energy can be traded with different prices based on the attributes of its source.

In the second thread, the network constraints have been considered in energy trading. Technical aspects of energy trading in a microgrid are addressed in [22] by introducing two indices to allocate attribution of power losses to each transaction. A distributed electricity trading system using multi-agent system is presented in [23], where a local coordinator agent is responsible for energy allocation considering network constraints. Authors in [24] propose consensus-based energy management in smart grid considering transmission losses. In this work, a distributed method is developed to reach global price and global power balance, without considering line flow constraints and product differentiation. An exogenous approach to account for network constraints and attribution of network usage is proposed in [6], where the system operator imposes a price signal for grid cost allocation. An alternative method to specify share of each transaction in the line flow constraint is employing PTDF. The concept of using PTDF to evaluate impact of each transaction on the line flow is presented in [25]. However, these approaches did not consider bilateral energy trading with different prices in a market with multiple sellers and buyers. The RCI method for multi-bilateral energy trading allowing product differentiation is proposed by authors in [14] which contributes to slow convergence. This paper is seeking to complement the existing works by proposing a decentralised algorithm for bilateral energy trading, which allows product differentiation and incorporates line flow constraints.

III. P2P ENERGY TRADING PLATFORM

A. Market properties

In the proposed market platform, all market players are assumed to be able to determine their level of generation and demand for a particular time slot. Market players can sell their surplus energy to or buy their deficient energy from the retailer. Alternatively, they can also trade energy among themselves via P2P energy trading considering the limited economic benefit from energy trading with the retailer. The proposed market structure is for a forward market, which can be interpreted as a resource allocation problem to dispatch generation and demand among players. In the P2P market, the agreements on bilateral trading are established according to the available energy.

The assumed energy network is same as the described network in [26], where the network is divided into physical layer and virtual layer. The physical layer includes electrical connection of all players, whereas the virtual layer is the communication network which can be used for information exchange among players. Different from wholesale market which has a single price for electricity, in the P2P market each transaction can have a different price to reflect players' preferences. Although product differentiation may result in sub-optimisation of overall energy price, the fact that all market players can negotiate simultaneously with each other can help prices to converge to similar values [14].

In this paper, a general model for market players is considered, which covers different types of market players including, small-scale consumers, industrial loads, and prosumers with storage systems and renewable generation. The market is defined independently of the nature of the market players' assets, which implies that different DERs can be included without the need to modify the market mechanism. Hence, market players are defined in the market by their net generation or demand which comes from the summation of generation or demand of all of their assets. The presented market is designed for active power trading, and the reactive power is assumed to be managed locally by the network provider. Losses due to power flows between market players are assumed to be relatively small and are not included in the model. Throughout this paper, emphasis is placed on a deterministic clearing mechanism for a single market time slot, which can be easily extended to multiple time slots. The time slot here is assumed to be one hour, and the terms energy and power are used interchangeably.

A market with $\mathcal{N}$ players composed of a set of $\mathcal{N}_S = \{1, \dots, N_S\}$ sellers and $\mathcal{N}_B = \{1, \dots, N_B\}$ buyers is considered. All sellers belong to $\mathcal{N}_S$ and all buyers belong to $\mathcal{N}_B$. Therefore, $\mathcal{N}_S \cup \mathcal{N}_B = \mathcal{N}$ and $\mathcal{N}_S \cap \mathcal{N}_B = \emptyset$. During each time interval, each individual subscriber behaves independently and tries to maximise its welfare. The objective function in the market is to maximise welfare of all market subscribers by maximising summation of all individual market players' welfare. The overall objective function for social welfare maximisation can be written as (1)

$$\max_{\boldsymbol{x},\boldsymbol{y}} \left(\sum_{i=1}^{N_S} WS_i + \sum_{j=1}^{N_B} WB_j \right) \tag{1}$$

where WS_i and WB_j are welfare of seller i and buyer j respectively. The optimisation objective is to find generation of all sellers ($\boldsymbol{x}$) and demand of all buyers ($\boldsymbol{y}$) to maximise social welfare. Thus, it is required to have a model for welfare as a function of energy in both seller side as well as buyer side.

B. Welfare on seller and buyer sides

Sellers in the market offer their surplus energy for trading with buyers. The cost of providing x_i energy offered by seller i to the market can be approximated by a quadratic convex function [27].

$$C_i(x_i) = \alpha_i x_i^2 + \beta_i x_i + \gamma_i \tag{2}$$

where α_i, β_i and γ_i are predetermined positive constants, which reflect the amount of energy that a seller is willing to sell at different prices at a given time. These parameters are unique for each seller and are dependent on the type of generation, load, and future risks of each seller. The welfare of the seller is modelled by (3)

$$WS_i = \sum_{j \in \mathcal{N}_B} \left(x_{ij} \lambda_{ij} \right) - C_i(x_i) \tag{3}$$

where x_{ij} and λ_{ij} are the traded energy and its corresponding price in transaction between seller i and buyer j. The sold energy by seller i in each time slot is the summation of any sold energy to all connected peers as in (4) and this energy is constrained by (5)

$$x_i = \sum_{j \in \mathcal{N}_B} x_{ij} \tag{4}$$

$$x_i^{min} \le x_i \le x_i^{max}, \qquad \forall i \in \mathcal{N}_S \tag{5}$$

where x_i^{min} and x_i^{max} are the minimum and maximum generation of seller i respectively. If the generation of seller is from renewable sources with zero marginal cost, it can be modelled as "must take" generation. The "must take" generation such as from solar or wind generation can be modelled by setting upper and lower limits equal to each other resulting in the corresponding x_i variables becoming constant instead of optimisation variable.

Buyers with flexible loads participate in the market assuming that their flexible loads are continuous in nature and they can adjust their demand regarding market price. The satisfaction level of the buyers is based on their demand from the market and can be modelled by utility function. The utility function is an important concept that measures preferences over a set of goods and services, and here it represents the welfare or satisfaction of a consumer from consuming a certain amount of energy. The common form of the utility function is a non-decreasing function with a non-increasing marginal benefit. Also, no energy consumption brings no benefit for buyers. Thus, the utility function for each buyer j with demand y_j should satisfies three properties:

$$\frac{dU_j(y_j)}{dy_j} \ge 0; \;\; \frac{d^2U_j(y_j)}{dy_j{}^2} \le 0; \;\; U_j(0) = 0. \tag{6}$$

To satisfy these properties, a quadratic utility function for buyer j is considered as proposed in [28].

$$U_j(y_j) = \begin{cases} \omega_j y_j - \delta_j y_j^2, & y_j < \omega_j / 2\delta_j \\ \omega_j^2 / 2\delta_j, & y_j \ge \omega_j / 2\delta_j \end{cases} \tag{7}$$

where ω_j and δ_j are predetermined positive constants and determine how a buyer response to different prices and y_j is total demanded energy by buyer j. As each buyer in the market behaves independently, these parameters vary among buyers and also at different times of a day or climate conditions. The

welfare of buyer j is defined as the utility of the demanded energy minus paid money for this energy as formulated in (8)

$$WB_j = U_j(y_j) - \sum_{i \in \mathcal{N}_S}(y_{ji}\lambda_{ij}). \tag{8}$$

Any buyer can demand energy from different sellers, and the total demanded energy is bounded by (10)

$$y_j = \sum_{i \in \mathcal{N}_S} y_{ji} \tag{9}$$

$$y_j^{min} \leq y_j \leq y_j^{max}, \qquad \forall j \in \mathcal{N}_B \tag{10}$$

where y_j^{min} denotes the baseline demand of buyer j, which is from "must-run" appliances and y_j^{max} is the maximum energy that buyer is willing to demand from the market. Buyers with inflexible loads can be modelled by setting upper and lower limits equal to each other, resulting in the corresponding y_j variable becoming constant instead of optimisation variable. It should be noted that various types of utility and cost functions can be used to model seller's cost and buyer's utility in the local energy trading. For the utility function, as long as it satisfies properties in equation (6), it can be used in the proposed method. The quadratic utility and cost function are more common in the literature as their derivative (marginal cost/profit) are linear which is more desirable in the optimisation problem. However, all the considerations in this paper hold as long as the entire optimisation problem is convex. The quadratic cost and utility functions are realistic for a large class of conventional generators, and indeed some of the best assumption to make when having limited insight in actual utility functions of small consumers and prosumers [14].

C. Optimisation problem

The objective function in this market is to maximise welfare of all players. Summation of (3) and (8) for all players gives:

$$\max_{\boldsymbol{x},\boldsymbol{y}}\left(\sum_{j=1}^{N_B} U_j(y_j) - \sum_{i=1}^{N_S} C_i(x_i)\right). \tag{11}$$

Demand-supply constraints force that the proposed energy by sellers and buyers in each bilateral trading should be equal, i.e.

$$x_{ij} = y_{ji}, \qquad \forall i \in \mathcal{N}_S, \forall j \in \mathcal{N}_B. \tag{12}$$

Players in the local market use the traditional grid to transfer energy and hence excluding traditional grid from local energy trading is infeasible. Therefore, both economic (social welfare) and technical (network constraints) aspects should be considered. The economics of the market subscribers is modelled by considering their welfare in the market. Here, line flow constraints are added as a constraint of the objective function to model physical network in the energy trading. The flow in each line is bounded by (13)

$$-F_\ell^{max} \leq F_\ell \leq F_\ell^{max}, \ \forall \ell \in \mathcal{N}_\ell. \tag{13}$$

where F_ℓ^{max} is the maximum capacity of line ℓ. The coupling between market and network constraints implies an intense involvement of the system operator in the market clearing, which needs a central entity for line flow calculation. A line utilisation charge can be considered to dissociate the market from network constraints and avoid having a central entity for power flow calculation. These price signals are only used when line flow constraints are disrespected. In this paper, PTDF is used to calculate flow of lines and indicate lines that are used for power transfer in each individual transaction. PTDF is defined as "linear approximation of the first order sensitivity of the active power flow and represents change in active power flow over certain line, caused by change in active power generation in certain node of electric power system" [29]. PTDF for line ℓ is shown by φ_{ij}^ℓ and indicates part of the generated energy by seller i which is transferred to buyer j through line ℓ and can be obtained using (14)

$$\varphi_{ij}^\ell = \psi_i^\ell - \psi_j^\ell \tag{14}$$

where ψ_i^ℓ , ψ_j^ℓ are Injection Shift Factors (ISF) in line ℓ for node i and j respectively. ISF is an approximation of the sensitivity matrix and quantifies the re-distribution of power through each branch following a change in generation or load in a particular node. The ISF matrix is shown by $\Psi \triangleq [\psi_n^\ell] i \epsilon \mathbb{R}^{N_\ell \times N_b}$, where N_b is the number of nodes in the system. This matrix can be obtained using (15) by having diagonal branch susceptance matrix (B'), branch to node incidence matrix (A), and reduced nodal susceptance matrix (C). In the matrix A, a_ℓ^T is ℓ^{th} row, in which a line exists between node n and m.

$$\Psi \triangleq B'AC^{-1} \tag{15}$$

$$B' \triangleq diag[b_1, b_2, \dots, b_L] \epsilon \mathbb{R}^{N_\ell \times N_\ell} \tag{16}$$

$$\begin{aligned} A &\triangleq [a_1, a_2, \dots, a_L]^T \epsilon \mathbb{R}^{N_\ell \times N_b} \\ a_\ell^T &\triangleq [0 \ \cdots 0 \ \overset{n}{1} \ 0 \cdots 0 \ \overset{m}{-1} \ 0 \cdots 0] \end{aligned} \tag{17}$$

$$C \triangleq A^T B' A \epsilon \mathbb{R}^{N_b \times N_b}. \tag{18}$$

By having PTDF and traded energy between sellers and buyers (x_{ij} and y_{ji}), the power flow in line ℓ can be obtained by (19)

$$F_\ell = \sum_{i=1}^{N_S}\sum_{j=1}^{N_B} \varphi_{ij}^\ell x_{ij} = \sum_{i=1}^{N_S}\sum_{j=1}^{N_B} \varphi_{ij}^\ell y_{ji} \ . \tag{19}$$

As at the convergence point $x_{ij} = y_{ji}$, both of them can be used in the line flow calculation. Once line flow is calculated, if this flow is lower or higher than the threshold, price signals $(\overline{\vartheta}_\ell, \underline{\vartheta}_\ell)$ will be sent to the players who are using that particular line to avoid overflow or congestion in the line. It is noteworthy that for any given network, PTDF matrix depends on the network structure and is independent of the power flows through the network. As the network structure does not change during energy trading, this matrix is a fixed matrix and can be calculated before the market clearing process. Hence, for each line in the network, players who need to use it for their transactions can be determined using PTDF matrix.

IV. Market Clearing Algorithm

In this section, a decentralised optimisation algorithm for solving (11) is proposed. The problem in (11) is a convex problem and should be maximised subject to (5), (10), (12) and (13). This problem can be solved in a centralised manner which needs a central controller who is aware of all market players' characteristics. Implementing the optimisation in a centralised manner endangers the privacy of the players as it needs to have all players' information to solve the problem. However, by decomposing the optimisation problem, the set of information needed to grant market efficiency can be limited to enhance the privacy level of the participants' utility and cost curves as well as their preferences. In the decentralised optimisation, all data and related computation are handled locally by each agent. The other incentive in implementing decentralised optimisation is to ensure that a failure of a single computational entity, i.e., the central coordinator, will not lead to an inability to control the system. Therefore, this paper aims at clearing the market in a decentralised manner without requiring a single computation centre or individual information of market players.

The optimisation problem can be split into several local subproblems, where each subproblem is locally solvable by players with limited information from connected players. Before splitting the global problem to subproblems, global and local constraints should be decoupled by applying Lagrangian multipliers. These multipliers are defined as follows to decouple constraints

$$\begin{aligned}&\mathcal{L}\left(x_{ij}, y_{ji}, \lambda_{ij}, \overline{\vartheta}_\ell, \underline{\vartheta}_\ell, \overline{\mu}_i, \underline{\mu}_i, \overline{\mu}_j, \underline{\mu}_j\right) = \\ &\sum_{j=1}^{N_B} U_j(y_j) - \sum_{i=1}^{N_S} C_i(x_i) + \sum_{i=1}^{N_S}\sum_{j=1}^{N_B} \lambda_{ij}(x_{ij} - y_{ji}) \\ &- \sum_{\ell=1}^{N_\ell} \overline{\vartheta}_\ell (F_\ell - F_\ell^{max}) + \sum_{\ell=1}^{N_\ell} \underline{\vartheta}_\ell (F_\ell + F_\ell^{max}) \\ &- \sum_{i=1}^{N_S} \overline{\mu}_i (x_i - x_i^{max}) - \sum_{i=1}^{N_S} \underline{\mu}_i (x_i^{min} - x_i) \\ &- \sum_{j=1}^{N_B} \overline{\mu}_j (y_j - y_j^{max}) - \sum_{j=1}^{N_B} \underline{\mu}_j (y_j^{min} - y_j)\end{aligned} \tag{20}$$

where $\lambda_{ij}, \overline{\vartheta}_\ell, \underline{\vartheta}_\ell, \overline{\mu}_i, \underline{\mu}_i, \overline{\mu}_j$, and $\underline{\mu}_j$ represent the Lagrangian multipliers associated with each of the constraints in (5), (10), (12), and (13) respectively. By taking derivative of $\mathcal{L}$ with respect to these multipliers, the distributed updating rules can be derived utilising the primal-dual gradient method [12]

$$\lambda_{ij}^{k+1} = [\lambda_{ij}^k - \xi_\lambda^k (x_{ij}^k - y_{ji}^k)]^+ \tag{21}$$

$$\overline{\mu}_i^{k+1} = [\overline{\mu}_i^k + \xi_\mu^k (x_i{}^k - x_i^{max})]^+ \tag{22}$$

$$\underline{\mu}_i^{k+1} = [\underline{\mu}_i^k + \xi_\mu^k (x_i^{min} - x_i{}^k)]^+ \tag{23}$$

$$\overline{\mu}_j^{k+1} = [\overline{\mu}_j^k + \xi_\mu^k (y_{ji}^k - y_j^{max})]^+ \tag{24}$$

$$\underline{\mu}_j^{k+1} = [\underline{\mu}_j^k + \xi_\mu^k (y_j^{min} - y_{ji}^k)]^+ \tag{25}$$

$$\overline{\vartheta}_\ell^{k+1} = [\overline{\vartheta}_\ell^k + \xi_\vartheta^k (F_\ell^k - F_\ell^{max})]^+ \tag{26}$$

$$\underline{\vartheta}_\ell^{k+1} = [\underline{\vartheta}_\ell^k - \xi_\vartheta^k (F_\ell^k + F_\ell^{max})]^+ \tag{27}$$

$$x_i{}^{k+1} = \left[(f_i')^{-1}(\lambda_{ij}^k)\right]_{x_i^{min}}^{x_i^{max}} \tag{28}$$

$$\begin{aligned}f_i(\lambda_{ij}) \triangleq &\sum_j (\lambda_{ij} x_{ij}) - C_i(x_i) \\ &- \sum_{j=1}^{N_B}\sum_{\ell=1}^{N_\ell} \varphi_{ij}^\ell (\overline{\vartheta}_\ell - \underline{\vartheta}_\ell) x_{ij}\end{aligned} \tag{29}$$

$$y_j{}^{k+1} = \left[\left(g_j'\right)^{-1}(\lambda_{ij}^k)\right]_{y_j^{min}}^{y_j^{max}} \tag{30}$$

$$\begin{aligned}g_j(\lambda_{ij}) \triangleq &U_j(y_j) - \sum_i (\lambda_{ij} y_{ji}) \\ &- \sum_{i=1}^{N_S}\sum_{\ell=1}^{N_\ell} \varphi_{ij}^\ell (\overline{\vartheta}_\ell - \underline{\vartheta}_\ell) y_{ji}\end{aligned} \tag{31}$$

where, ξ_λ^k, ξ_μ^k, and ξ_ϑ^k are tuning parameters, $[.]^+$ denotes the max(.,0) and $[.]_a^b$ denotes min{max(.,a), b}. Equations (21) to (27) involve updating of Lagrangian multipliers, whereas (28) and (30) are the local problems of seller and buyer respectively. These local problems can be solved directly by maximising (29) and (31) using methods such as interior points. However, solving of these local problems can be computationally costly due to the high number of decision variables per agent. Also, it increases the computation time in each iteration, which makes the convergence rate slower. Hence, instead of solving (29) and (31), they can be replaced with a gradient step and projection on the feasibility system. Compared to interior-point methods, first-order methods need more iterations for convergence. However, due to simplicity of computations per iteration, these methods have lower computation time and are particularly effective for large-scale optimisation problem [30]. The first order optimality for all sellers and buyers yields that:

Algorithm 1 Market clearing algorithm

```
Initialisation initialise variables λ, x, y, μ, ϑ, φ
while convergence criteria are not satisfied do
    for all i ∈ N_S (in parallel)
        update λ using (21)
        update μ̄_i and μ_i using (22) and (23)
        update x using (36)
        calculate stopping criteria using (39)
        communicate λ to buyers and x to lines
    end
    for all j ∈ N_B (in parallel)
        update μ̄_j and μ_j using (24) and (25)
        update y using (35)
        calculate stopping criteria using (40)
        communicate y to sellers and lines
    end
    for all ℓ ∈ N_ℓ (in parallel)
        calculate F_ℓ using (19)
        update ϑ̄_ℓ and ϑ_ℓ using (26) and (27)
        calculate stopping criteria using (41)
        if ϑ̄_ℓ, ϑ_ℓ ≠ 0
            communicate ϑ̄_ℓ and ϑ_ℓ to players
        end
    end
end
```

$$\begin{aligned}\nabla_{x_i}\mathcal{L} = &-2\alpha_i x_i - \beta_i + \sum_{j=1}^{N_B} \lambda_{ij} \\ &- \sum_{j=1}^{N_B}\sum_{\ell=1}^{N_\ell} \varphi_{ij}^\ell \left(\overline{\vartheta}_\ell - \underline{\vartheta}_\ell\right) - \overline{\mu}_i + \underline{\mu}_i = 0, \forall i \in \mathcal{N}_S\end{aligned} \tag{32}$$

$$\begin{aligned}\nabla_{y_j}\mathcal{L} = &\omega_j - 2\delta_j y_j - \sum_{i=1}^{N_S} \lambda_{ij} \\ &- \sum_{i=1}^{N_S}\sum_{\ell=1}^{N_\ell} \varphi_{ij}^\ell \left(\overline{\vartheta}_\ell - \underline{\vartheta}_\ell\right) - \overline{\mu}_j + \underline{\mu}_j = 0, \forall j \in \mathcal{N}_B.\end{aligned} \tag{33}$$

The Lagrangian of the relaxed problem has bijective gradient when seen as function of the power setpoints. For each trade between seller i and buyer j, the target energy setpoint can be defined using the inverse gradient as:

$$\hat{x}_{ij}^{k+1} = \frac{\lambda_{ij}^{k+1} - \sum_{\ell=1}^{N_\ell} \varphi_{ij}^\ell (\overline{\vartheta}_\ell - \underline{\vartheta}_\ell) - \overline{\mu}_i^{k+1} + \underline{\mu}_i^{k+1} - \beta_i}{2\alpha_i} \tag{34}$$

$$\hat{y}_{ij}^{k+1} = \frac{\omega_j - \lambda_{ij}^{k+1} - \sum_{\ell=1}^{N_\ell} \varphi_{ij}^\ell (\overline{\vartheta}_\ell - \underline{\vartheta}_\ell) - \overline{\mu}_j^{k+1} + \underline{\mu}_j^{k+1}}{2\delta_j}. \tag{35}$$

The primal variables are then updated using (36) and (37)

$$x_{ij}^{k+1} = [x_{ij}^k + \mathscr{f}_{ij}^k (\hat{x}_{ij}^{k+1} - x_i^k)]^+ \tag{36}$$

$$y_{ji}^{k+1} = [y_{ji}^k + \mathscr{f}_{ji}^k (\hat{y}_{ij}^{k+1} - y_j^k)]^+ \tag{37}$$

where, $\mathscr{f}_{ij}^k$ and $\mathscr{f}_{ji}^k$ are asymptotically proportional factors defined as:

$$\mathscr{f}_{ij}^k = \frac{|x_{ij}^k| + \upsilon^k}{\sum_{j=1}^{N_B}(|x_{ij}^k| + \upsilon^k)}, \mathscr{f}_{ji}^k = \frac{|y_{ji}^k| + \upsilon^k}{\sum_{i=1}^{N_S}(|y_{ji}^k| + \upsilon^k)} \tag{38}$$

where υ^k is a positive and persistent sequence. The iterative algorithm for market clearing is demonstrated in **Algorithm 1** and Fig. 1 illustrates the information flow for a simple market with one seller, two buyers, and three lines. Before starting market clearing process, market players will manage their assets to indicate minimum and maximum of their

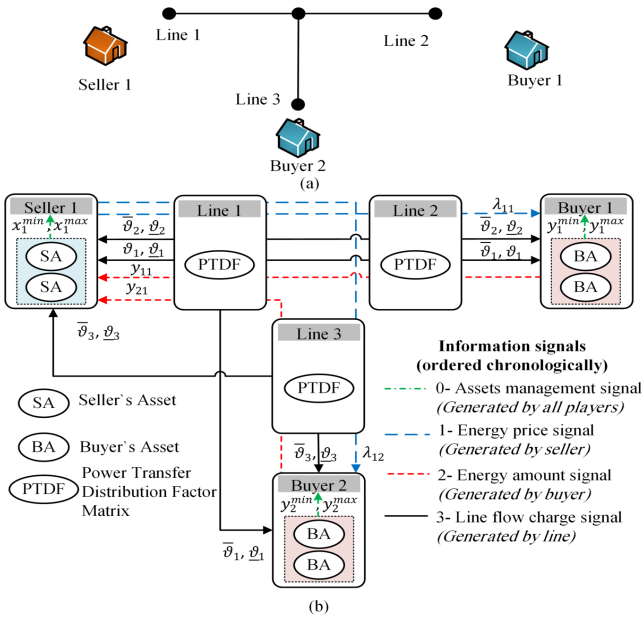

Fig. 1. A simple network for information flow demonstration; (a) Physical layer (b) Virtual layer

generation/demand. At the first step of market clearing, all sellers in parallel will update their prices for different buyers using (21). Since each transaction can have a different price, the product differentiation is implemented in the market. Then, energy for each seller is updated using (36). The updated prices by sellers will be sent to the buyers, where they will update their energy using (37). The line flow constraints are modelled in the optimisation using two Lagrangian multipliers which act as penalty factors. Flow in each line is calculated using (19), and line flow prices are updated using (26) and (27). Then, line agents will send line flow prices to the corresponding players who are using the line for energy trading. For each individual line, players who should receive price signals are determined using PTDF matrix. In each iteration, when players are updating their demand and generation, they will consider these prices to increase their welfare. Due to applying these price signals for line utilisation, players will try to trade energy with nearby players, which reduces power losses and justifies the assumption of neglecting these losses. Each agent will repeat its algorithm to meet stopping criteria. Stopping criteria for sellers, buyers, and lines are as follow respectively

$$\left|\lambda_{ij}^{k+1}-\lambda_{ij}^{k}\right|<\varepsilon_{\lambda},\ \left|\mu_{i}^{k+1}-\mu_{i}^{k}\right|<\varepsilon_{\mu} \tag{39}$$

$$\left|\mu_{j}^{k+1}-\mu_{j}^{k}\right|<\varepsilon_{\mu} \tag{40}$$

$$\left|\vartheta_{\ell}^{k+1}-\vartheta_{\ell}^{k}\right|<\varepsilon_{\vartheta} \tag{41}$$

where ε_{λ}, ε_{μ}, and ε_{ϑ} are small positive numbers. Although the proposed approach is sequential, it is different from ADMM as it does not require a global update of the Lagrange multiplier associated with the power balance. Instead of generating a global update of Lagrange multiplier, a set of multipliers associated with each individual trade is defined and updating rules are assigned to individual market players.

In the presented decentralised energy trading scheme, each player is supposed to be a rational agent who takes the most beneficial decisions and tries to maximise its own welfare. This means that in practice, through the decentralised solution and by decomposing the objective function, all involved players focus only on solving their own local welfare maximisation problem, which contributes to social welfare maximisation. Social welfare maximisation can maximise the total welfare of the market players in the electricity market considering that each individual player profit is maximised too [31].

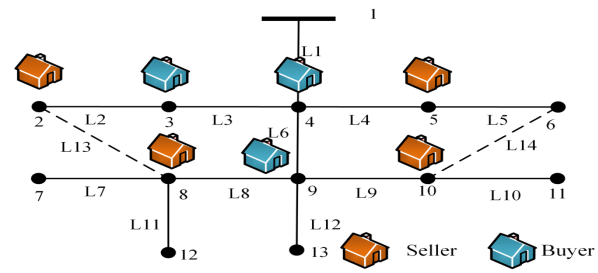

Fig. 2 Test system schematic (Solid line: radial network- Dashed line: Meshed network)

TABLE I

INPUT DATA; SELLERS AND BUYERS PARAMETERS

Sellers parameters					
Index	Node	α_i (¢/kWh)	β_i (¢/kWh²)	x_i^{min} (kW)	x_i^{max} (kW)
1	2	0.084	5.21	0	6.77
2	5	0.075	6.35	0	5.17
3	8	0.051	7.98	0	4.42
4	10	0.079	8.25	0	2.57
Buyers parameters					
Index	Node	ω_j (¢/kWh)	δ_j (¢/kWh²)	y_j^{min} (kW)	y_j^{max} (kW)
1	3	0.072	8.21	0	7.88
2	4	0.084	11.21	0	5.14
3	9	0.045	13.24	0	3.07

V. NUMERICAL RESULTS

This section provides numerical results for performance evaluation of the proposed decentralised approach through different case studies. Case studies were run on a computer with an Intel Core i7 processor running at 2.60 GHz using 16 GB of RAM.

A. Simulation setup

For ease of illustration, in the first case study, a small residential distribution network with seven players (4 sellers and 3 buyers) is considered. The considered test system is a radial 13 nodes network derived from IEEE 13 nodes distribution test feeder. As shown in Fig. 2, each agent is connected to one node and sellers are located at nodes 2, 5, 8, and 10 and buyers are placed at nodes 3, 4, and 9. The connections indicate the physical electrical connections, and for communication network, it is assumed that there exists a connected network for all agents to negotiate. Sellers and buyers parameters are considered as shown in Table I. Emphasis is eventually placed on a deterministic clearing mechanism for a single market time slot. The proposed method ensures convergence of the algorithm using any set of tuning parameters. However, the convergence rate and number of iterations are dependent on the tuning parameters and can be changed by adjusting these parameters. The tuning parameters are set to $\xi_{\lambda}^{k}=\frac{0.1}{k^{0.009}}$, $\xi_{\mu}^{k}=\xi_{\vartheta}^{k}=0.01$, and $\upsilon^{k}=\frac{1}{k^{0.01}}$. All stopping criteria defined in (39) to (41) are set to 0.001.

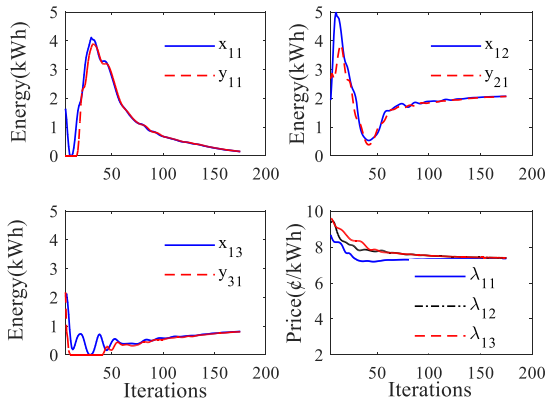

Fig. 3. Evolution of traded energy and perceived prices by seller 1

TABLE II
WELFARE OF PLAYERS IN TRADING WITH RETAILER AND P2P MARKET

Welfare (¢)	WS_1	WS_2	WS_3	WS_4	WB_1	WB_2	WB_3	Total
Trading with retailer	1.51	-3.81	0	0	-59.34	-67.06	-34.45	-163.15
P2P market	13.09	5.82	0	0	1.31	14.43	14.59	56.24

B. Optimality and convergence

The evolution of the traded energy among seller 1 and all three buyers and its corresponding clearing price are shown in Fig. 3, where it can be verified that in each bilateral trading between seller and buyer they will reach convergence. Also, the final allocation satisfies seller's constraints. In this case study, as there is no overflowed line, the final clearing price for all buyers is the same. However, if a seller has to pay a line utilisation charge for selling energy to a specific buyer, the clearing price for that buyer would be higher, which motivates players to trade energy locally to reduce line congestion and power losses.

To demonstrate how the proposed P2P trading scheme can help players to increase their welfare, the welfare of players in the proposed P2P market is compared with their welfare in traditional method of trading with retailers. The values of the retailer's selling price and Feed-in Tariff (FiT) price are assumed to be 24.6 and 6 ¢/kWh respectively. The results are tabulated in Table II, and verify that players' welfare is increased by participating in the proposed P2P market. It should be noted that market players are aware of other prices such as retailer's selling price and FiT and will participate in the P2P market taking into account the external markets as a backup. It means that a seller cannot offer a price higher than retailer's selling price and a buyer cannot demand energy at a price lower than FiT. As long as the prices in the P2P market are lower than market price and higher than FiT, players can increase their welfare by participating in the P2P market.

To evaluate optimality and convergence of the proposed approach, this method has been compared with other methods which can be applied for market clearing in P2P markets. The results are tabulated in Table III. The first applied method is an ADMM-based distributed market clearing method, as proposed in [15]. In this method, there is a central coordinator who receives amount of generated/demanded energy from sellers/buyers and iteratively calculates the clearing price. The other method is the RCI method introduced in [14], where each player sends amount of energy and price to all connected players for direct negotiation. Also, the proposed algorithm in this paper is implemented in two ways; using interior-point method to solve (28) and (30), where MATLAB in-built function *fmincon* is employed, and using first-order method by applying direct gradient method to solve players' local problems in (36) and (37). Finally, a centralised optimisation is implemented to solve (11) as a convex optimisation problem, by having all information from all players.

TABLE III
QUANTITATIVE AND QUALITATIVE COMPARISON OF THE PROPOSED METHOD WITH OTHER MARKET CLEARING METHODS

	Distributed				Centralised
	ADMM	Proposed Method Interior-point	Proposed Method First-order	RCI	
Welfare (¢)	50.75	50.35	49.24	49.64	50.75
Traded energy (kWh)	11.89	11.89	11.85	11.86	11.89
No. of iteration	55	139	176	497	-
Convergence time (s)	1.74	111.5	2.56	16.36	-
Properties					
Privacy of players	✓	✓	✓	✓	×
Decentralised	×	✓	✓	✓	×
Product differentiation	×	✓	✓	✓	×
Low shared information	✓	✓✓	✓✓	✓	×

The results in Table III verify that the proposed method can reach approximately the optimal outcomes obtained from centralised optimisation. Also, the social welfare and total traded power obtained from the proposed method are similar to the results from other distributed methods, which confirms the optimality. It should be noted that the proposed algorithm needs lower time and fewer iterations than RCI to converge. Reduction in the number of iterations and convergence time is due to decrease in the exchanged information among sellers and buyers. In the proposed method, instead of sending both energy and price, each seller only sends price to the buyers, and each buyer only sends demanded energy to the sellers.

Properties of the implemented methods are compared in the second half of Table III. An essential feature of any competitive market is the equal opportunity for competition of all players. Therefore, the privacy of information of players should be secured. In the distributed methods, each player can solve its problem locally without any need to reveal the private information, but in the centralised method, all information from players should be gathered in one centre. One of the main incentives to use P2P market is the possibility of having product differentiation in energy trading, where different prices can be considered for each and every transaction. The product differentiation allows reflecting consumers' preferences. Among the distributed methods for market clearing, the implemented ADMM method is not decentralised, as it needs to have a central coordinator to calculate and generate coordination signal. Having a central coordinator can reduce the convergence time and number of required iterations, but it endangers the scalability of the system. Also, implementing product differentiation in the ADMM method is not straightforward. As discussed before, compared to RCI method,

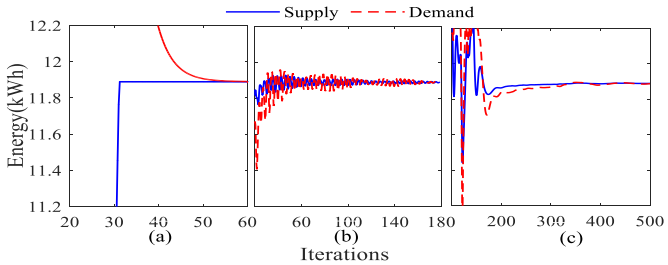

Fig. 4. Convergence of total demand and supply in different algorithms: (a) ADMM, (b) Proposed method, (c) RCI

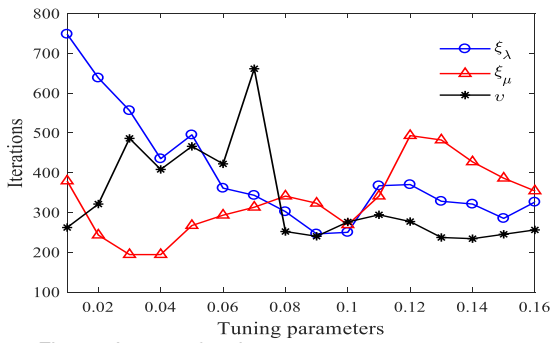

Fig. 5. Impact of tuning parameters on convergence

in the proposed approach a lower amount of information should be shared, which reduces communication overheads.

The convergence of total demand and supply in different methods is shown in Fig. 4, which verifies that the total traded energy in all of these methods is the same. Fig. 4 (a) illustrates the convergence in the ADMM method. In this method, as there is one unit price for all players, the convergence is faster. By increase in the market price, sellers increase their generation and buyers decrease their demand till total demand and supply are converged. Results for the proposed method and the RCI method are shown in Fig. 4 (b) and (c) respectively. In these methods there is no unit price for the market and each transaction can have a different price. As each player can negotiate with other players directly, more iterations are needed for the convergence. However, in the proposed algorithm, the exchanged information in each transaction is lower which contributes to the lower number of iterations in comparison with the RCI method.

As stated earlier, compared with Consensus + Innovation method the proposed method has lower sensitivity to the tuning parameters and reaching convergence to the optimal solution can be guaranteed using a range of different tuning parameters. As specified in [13], to ensure convergence in Consensus + Innovation method, tuning parameters should be decaying as iteration index is increasing. Despite Consensus + Innovation method, the proposed method in this paper reaches the optimal equilibrium point even with constant tuning parameters for sufficiently small values. However, the number of iterations is different for each tuning parameter. A case study is performed to demonstrate the robustness of the proposed method, where the tuning parameters ξ_λ^k, ξ_μ^k, and υ^k are changed and number of iteration for convergence is observed. Results are shown in Fig. 5, which demonstrate that for different values of tuning parameters the algorithm always will converge to the optimal solution, though with different number of iterations. It should be noted that in this case study tuning parameters are assumed

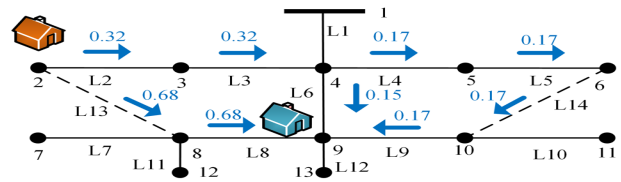

Fig. 6. PTDF calculation for meshed network (transaction between seller 1 and buyer 3)

TABLE IV
IMPACT OF NETWORK CONFIGURATION ON ALGORITHM PERFORMANCE

Network configuration	Welfare (¢)	Traded energy (kWh)	No. of iteration	Convergence time (s)
Radial	49.24	11.85	176	2.56
Meshed	49.23	11.83	**245**	**6.45**

to be constant for all iterations. The number of iterations can be reduced by adjusting tuning parameters in decaying form, as depicted in the first case study.

C. Effects of network configuration

Impact of network configuration on the performance of the proposed mechanism is studied in this section. PTDF matrix for both radial and meshed network can be calculated using equations (14) to (18). Changing network configuration by adding new lines will change the dimension of matrices Ψ, B' and A and consequently PTDF matrix. Generally, in the radial networks there is only one path for transferring power from a seller to a buyer, while in the meshed network different paths exist to transfer power from one node to another node. In this section, transaction between seller 1 and buyer 3 is used as an example to demonstrate impact of network configuration on PTDF for different lines. Since PTDF is a linear approximation of power flow, the contribution of different transactions can be determined with superposition. As it can be seen from Fig. 2, in the radial network there is only one path for transferring energy from seller 1 to buyer 3, and only lines 2, 3, and 4 are used in this transaction. The presented test system in Fig. 2 is changed to a meshed configuration, by adding two additional lines (connecting nodes 2 to 8, and node 6 to 10). As illustrated in Fig. 6, in the meshed network there are different paths for transferring power and lines 2, 3, 4, 5, 6, 8, 9, 13, and 14 are used for this transaction. Arrows and their associated numbers indicate the PTDF for different lines due to transferring power from seller 1 to buyer 3. In the meshed networks there are different paths for power flows and hence, number of lines which are being used for a transaction is more than radial network and more communication is needed between market players and line agents.

The proposed method is tested for the meshed network to show impact of network configuration on the algorithm performance. Results for the meshed network are compared with the radial network as presented in Table IV, which verify that the proposed method can be applied to both radial and meshed networks and can reach approximately the same results. However, the required time and number of iterations for convergence in the meshed network would be higher, which is due the fact that in the meshed network number of players who use a particular line is significant, and the line agent needs to communicate with more players. Hence, implementing this method for the meshed network requires more communication and computation overheads.

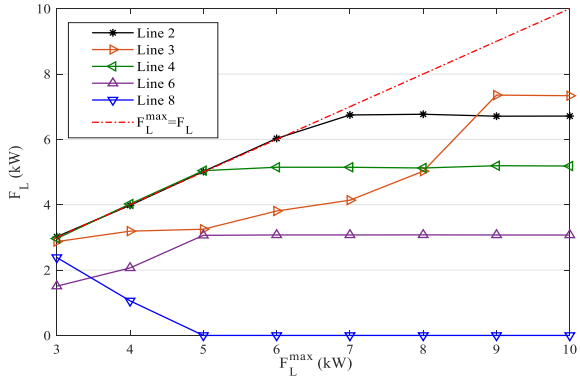

Fig. 7. Power flow in different lines for different line capacities

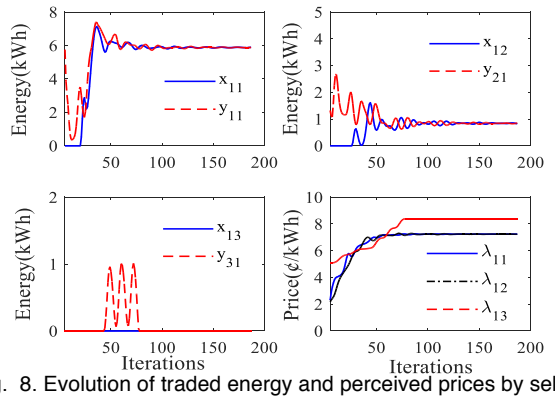

Fig. 8. Evolution of traded energy and perceived prices by seller 1 after congestion in line 6

D. Line congestion management

In order to verify performance of the proposed method in the line congestion reduction, a DC power flow is performed to extract flow in different lines of the radial network for different line capacities. In the considered test system, the power flow in lines 2, 3, 4, 6, and 8 are non-zero, and results are shown for these lines only. The maximum line capacity for these lines is changed from 3 to 10 kW. The results are illustrated in Fig. 7. and confirm that the power flow in these lines is always lower than or equal to the maximum line capacity, which means that the proposed algorithm can control line flow constraints in bilateral contracts. In the case that there is a congested line in the system, players will avoid transactions that need to use the congested line. This is demonstrated in a case study, where it is assumed that line number 6 in Fig. 2 (the connecting line between nodes 4 and 9) is congested, and players who are using this line have to pay an additional charge. The evolution of traded energy by seller 1, in this case, is illustrated in Fig. 8. Comparing results in Fig. 3 and 8 yields that, the traded energy between seller 1 and buyer 3 is reduced to zero since this transaction needs to use the congested line, which imposes additional cost to the seller and buyer and line utilisation charge for this transaction is higher than other transactions. Instead, seller tries to trade energy with near buyers to avoid using the congested line.

E. Scalability

In practical case studies, the P2P market will be deployed for a large number of market players and the number of transactions would be significant. Also, market players should decide on

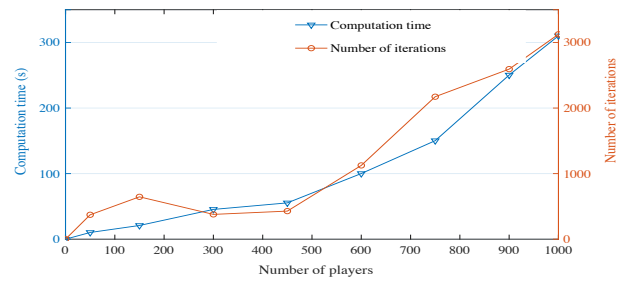

Fig. 9. Impact of number of players on computation time and required number of iteration for convergence

price and amount of traded energy at the beginning of each time slot. Therefore, the computation time of the algorithm and number of iterations are two critical factors affecting the scalability of the algorithm. To investigate the scalability issue, the proposed method is tested with higher number of players and different network configuration. The test network in Fig. 2 is augmented by extending the end-nodes and adding new players to the new nodes. Fig. 9 shows the impact of number of transactions in the market on the computation time of the proposed algorithm and number of iterations. Both of the computation time and number of iterations are increased by increasing the number of players. However, as the market clearing is performed for the next time slot, the computation time and number of iterations are still acceptable for large number of market players and verifies that the proposed method is implementable for a local market with large number of players.

It is worth mentioning that the above simulations are based on a fully connected communication network between all players, which means that each player can negotiate with all other players in the market. However, the scalability of the P2P markets can be enhanced by decreasing number of trading partners of each player. Different methods can be applied for sparsification of the communication matrix such as market segmentation and local market clearing. An example of market segmentation method is reported in [32], where an adaptive segmentation method is applied to reduce number of negotiating players in the P2P market to enhance the scalability of the market. The proposed method can be integrated with a market segmentation method by adding an additional step before starting bilateral energy trading process, where market players need to submit their offers/bids which will be used to cluster players to different segments. Then, the presented bilateral trading scheme can be applied for market clearing in each segment. As the number of players in each segment is fewer, the computation time and number of iterations for convergence would be lower.

VI. Conclusions

In this paper, a decentralised Peer-to-Peer (P2P) energy trading scheme is presented for direct interaction of sellers and buyers in the electricity market. In the proposed method, market players participate in the forward market and try to maximise their welfare through bilateral energy trading. A novel algorithm using primal-dual gradient method is developed to clear the market in a fully decentralised manner. Instead of interior-point method, first-order method is employed to solve

local optimisation problems, which reduces convergence time of the algorithm. Line flow constraints are modelled in the bilateral energy trading by defining price signals to avoid overflowed or congested lines in the system. This market structure respects market players' preferences by allowing bilateral energy trading with product differentiation. Implemented case studies verify that the proposed method can maximise market players' welfare and can reach the same optimal results as other distributed methods, while it needs lower information exchange. Also, applying line utilisation charge ensures that line flow constraints are respected in any individual transaction. The proposed method can be easily implemented, as there is no need to have a central coordinator and each player needs only to share a limited amount of information.

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Mohsen Khorasany (S'10) received the B.Sc. degree from Shahid Beheshti University, Tehran, Iran and M.Sc. degree from Shiraz University, Fars, Iran in 2010 and 2012, respectively. He is currently a Ph.D. student at Queensland University of Technology. His research interests include distributed energy resources, electricity market, and distributed optimization.

Yateendra Mishra (S'06, M'09) received the Ph.D. degree from the University of Queensland, Brisbane, Australia, in 2009. He is currently a senior lecturer and advanced QLD research fellow at the School of Electrical Engineering and Computer Science, Queensland University of Technology, Brisbane QLD, Australia. His research interests include distributed generation and distributed energy storage, and power system stability and control and their applications in smart grid.

Gerard Ledwich (SM'89) received the Ph.D. degree in electrical engineering from the University of Newcastle, Newcastle, Australia, in 1976. He is a Chair Professor in Electrical Asset Management at the School of Electrical Engineering and Computer Science, Queensland University of Technology, Brisbane QLD, Australia. His research interests are in the areas of power systems, power electronics, and wide area control of smart grid.